Weather Impact on the Operations of Utrecht Centraal Station

Group 20 - research proposal

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1 Research Proposal

From the night of January 6th to January 8th, a snowstorm hit The Netherlands, majorly affecting the rail network. This project aims to look at the duration of train delays, the number of cancellations and punctuality of the NS train operator during the snowstorm.

With the following research question:
How did the snowstorm of 6 January 2026 affect the duration of train delays, the number of cancellations, and punctuality at Utrecht Centraal, compared with a typical day under normal weather conditions?

The following objectives have been set:

1. To establish a baseline of NS operations and performance at Utrecht Centraal on a typical day under normal weather conditions, in terms of delay duration, number of cancellations, and punctuality.
2. To describe NS operations and performance at Utrecht Centraal on the 6th of January 2026, during the snowstorm, using the same indicators.
3. To compare the baseline with the snowstorm day in order to determine the impact of the snowstorm on delay duration, number of cancellations, and punctuality.

1.1 Data Obtained

The datasets used in this project are from KNMI [1] and Rijden de Treinen [3]. The first dataset contains detailed hourly records of weather records per day at De Bilt weather station, 4 km northeast of Utrecht Centraal, including:

- Minimum temperature
- Maximum temperature
- Precipitation amount and duration
- Weather code: rainfall, snowfall, thunder, icing
- Maximum hourly precipitation

The second dataset contains detailed records of train arrivals, departures, delays and cancellations, including:

- Service type (e.g. Sprinter, InterCity)
- Cancellation type (completely, partly)
- Maximum delay for a certain service
- Arrival delay
- Departure delay
- Arrival or departure canceled

Using the resources mentioned above, day-level data are retrieved in CSV format and converted into a structured pandas dataframe.

1.2 Research Questions

To address the research question, the following sub-questions are established:

- What does a typical day of operations for NS look like at Utrecht Centraal under normal weather conditions, in terms of delay duration, number of cancellations, and punctuality?
- What were the operations and performance of NS at Utrecht Centraal on the 6th of January 2026, during the snowstorm, in terms of delay duration, number of cancellations, and punctuality?
- How did delay duration, number of cancellations, and punctuality at Utrecht Centraal on the snowstorm day differ from the baseline of a typical day?*

1.3 Methods and Pipeline

The general steps to create and interpret the model include:

- Data preparation: the datasets are cleaned and restricted to the studied period.
- Identification of snowstorm period in the data: The KNMI weather data and train dataset is complementary used to cross-check the affected snow period based on weather indicators.
- A baseline period representing typical NS operations at Utrecht Centraal is established using days in January with comparable operating conditions but without severe weather.
- Performance comparison: Train performance during the snowstorm is compared with the baseline using delay, cancellation and punctuality indicators.

1.4 Scale of the Study

The study on the effects of weather on NS operation entails a scope centered around a central major railway station in the Netherlands, Utrecht Centraal Station. This station is of particular interest due its centrality of operations, both in operational as well as in graph-form. The induced delays on the trains in and out of the station are already relatively high, being an easy bottleneck for knock-on delays due to the complexity of switches, high usage of current capacity and operational tracks [4][2].

The temporal scope is a period of 2 weeks at the beginning of January 2026 when most parts of Europe were affected by heavy snowfall.

References

1. KNMI: Uurgegevens van weerstations. https://uurgegevens.knmi.nl/klimatologie/uurgegevens (2026)
2. Oorsprong, J., Faber, J.: Station utrecht centraal in ontwikkeling (Jun 2021), https://koerstue.nl/koersief/online/previously-on-the-koersief/utrechtcs
3. Rijden de Treinen: Open data over treinritten. https://www.rijdendetreinen.nl/open-data/treinarchief (2026), open data over Dutch train services, delays and cancellations
4. Zegeling, J.: Impact assessment of rail switch location design on railway network resilience. Delft, The Netherlands (Jun 2024)